PREP 2, LESSONS 3–4

Lifting and stretching

Gravitational potential energy; stretching springs; spring constant

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard unit of energy. DN 5 [1]

Joule (J)

2 Name the four ways energy is transferred between stores. DN 16 [1]

Mechanically, electrically, by heating and by waves (or by radiation)

3 Name the two quantities that kinetic energy depends on. DN 643 [1]

Mass and speed

Key ideas LESSONS 3–4

Show your working

4 A 2.0 kg box is lifted onto a shelf 1.5 m high. Use $E_p = m\,g\,h$ and $g = 9.8$ N/kg.
Calculate the energy transferred to the gravitational potential store of the box. [3]

$E_p = m\,g\,h$ ✓

$E_p = 2.0 \times 9.8 \times 1.5$ ✓

$E_p = 29.4$ J (29 J) ✓

5 A spring is 12.0 cm long. A 3.0 N force stretches it to 18.0 cm. Use $F = k\,e$

a Calculate the extension in metres. [1]

18.0 − 12.0 = 6.0 cm = 0.060 m ✓

b Calculate the spring constant. [2]

$k = \frac{F}{e}$ ✓ (rearranged)

$k = \frac{3.0}{0.060} = 50$ N/m ✓

c Up to the limit of proportionality, extension is [1]

directly proportional | inversely proportional | not related to force.